

Fig. 6: An actual-size, single-side PCB for the robot

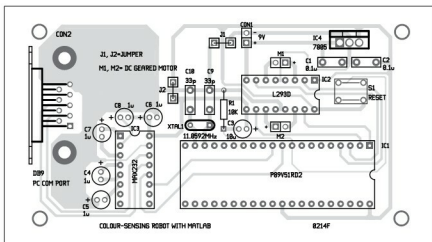


Fig. 7: Component layout for the PCB

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on source.efymag.com website.

all other pixels with the value 0 (black)
stats = *regionprops(diff_in,*
'BoundingBox','Centroid'). Measures
 a set of properties for each connected
 component (object) in the binary im-
 age. The image stats is a logical array
 and can have any dimension

By using 'regionprops()' function, we find the coordinates of centroid and bounding box as follows:

```
stats = regionprops(diff_im,  
    'BoundingBox', 'Centroid');  
for object = 1:length(stats)  
    bb = stats(object).BoundingBox
```

```
bc = stats(object).Centroid;
```

The 'Centroid' in the code specifies the centre of mass of the region. Note that the first element of centroid is the horizontal coordinate (or x-coordinate) of the centre of mass, and the second element is the vertical coordinate (or y-coordinate). Once we get the coordinates, we define conditions for the object's position and corresponding data to be sent to the robot as follows:

```
if (bc(1)>110 && bc(1)<170 && bc(2)
> 110 && bc(2) < 170)
```

We also get the idea of the current position of the object on GUI using the code below:

```
if (bc(1)>110 && bc(1)<170 && bc(2)
> 110 && bc(2) < 170)
axes(handles.axes5);
imshow('red2.jpg');
```

Fig. 4 indicates the position of the object in the defined centre range and Fig. 5 indicates that the object is raised to give a forward signal. The same is done for left, right, forward and backward movement.

Construction and testing

An actual-size, single-side PCB for the robot is shown in Fig. 6 and its component layout in Fig. 7. Assemble the components on the PCB and connect the motors and battery to build the robot. Use suitable bases for the ICs. Before inserting the MCU in the circuit burn the code into it using a suitable programmer. Connect the robot to the computer either through the serial port or USB port (using USB-to-serial converter) and check whether the USB-to-serial converter is detected or not in the device manager and change the COM port with respect to it.

Install MATLAB 7.10 or higher version in your system. Once it is installed, connect it to the robot. Also check which COM port you are using and write the same COM port in the code, otherwise you will get an error.

Run application

Once the setup is ready, follow the below-mentioned steps:

1. Download the required source code folder from this month's EFY DVD.
2. Run the program `gesture.m` and a GUI will appear as shown in Fig. 3.
3. Click on COM OPEN button to open the COM port.
4. Click on START and stand in front of the computer's camera or webcam with the red-coloured object in your hand.
5. Move the object up, down, left and right, and MATLAB program will send the data to the microcontroller according to the position of the object and the robot will move according to the values the microcontroller is receiving. ●

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